My formula sheet

One page that you write yourself as the course goes. Name every symbol in plain words, and write the industry term.

Linear algebra

Name	Formula	What each symbol means	Where it appears in AI
Dot product			
Cosine similarity			
Matrix multiplication			
Least squares (normal equation)			
Eigenvalue and eigenvector			
PCA			

Calculus and optimisation

Name	Formula	What each symbol means	Where it appears in AI
Derivative			
Gradient			
Chain rule			
Gradient descent update			
Backpropagation step			

Probability and statistics

Name	Formula	What each symbol means	Where it appears in AI
Expectation and variance			
Gaussian distribution			
Bayes' theorem			

Name	Formula	What each symbol means	Where it appears in AI
Maximum likelihood			

Information theory and attention

Name	Formula	What each symbol means	Where it appears in AI
Entropy and cross-entropy			
Softmax			
Attention			